

Power Distribution System Analysis

Using Gaussian Elimination for Real-World Engineering Problems

Advanced Engineering Mathematics Tutorial

May 11, 2025

From Linear Algebra to Practical Power Engineering

Key Engineering Concept

Executive Summary

This tutorial demonstrates how Gaussian elimination solves a real-world power distribution problem where three generators must supply exactly 100 kW while maintaining phase balance constraints. Through comprehensive visualizations and step-by-step analysis, we'll see how mathematical abstractions transform into practical engineering solutions.

1 The Power Distribution Challenge

Power System Example

Real-World Engineering Problem

Imagine you're managing a small power distribution network with three generators that must work together to supply exactly 100 kW to the grid. However, these generators also need to maintain phase balance to prevent grid instability. This creates a system of constraints that must be satisfied simultaneously:

1. **Total Power Constraint:** $P_1 + P_2 + P_3 = 100$ kW
2. **Phase Balance 1:** $0.8P_1 - 0.2P_2 - 0.1P_3 = 20$
3. **Phase Balance 2:** $0.1P_1 + 0.9P_2 - 0.3P_3 = 15$

Where P_1 , P_2 , and P_3 represent the power output in kilowatts from generators 1, 2, and 3 respectively.

Deep Insight

Why This Problem Matters

In power engineering, maintaining proper phase balance is crucial for:

- Preventing grid instability
- Minimizing power losses
- Ensuring efficient energy distribution
- Avoiding equipment damage from phase imbalances

The coefficients in our equations represent the phase contribution of each generator to the overall system balance—a simplified model of complex electrical engineering principles.

2 Mathematical Framework: Setting Up the System

2.1 Question 1: Formulating the Augmented Matrix

Step-by-Step Solution

Converting Equations to Matrix Form

Our system of equations becomes the augmented matrix:

$$[A|b] = \left[\begin{array}{ccc|c} 1.0 & 1.0 & 1.0 & 100.0 \\ 0.8 & -0.2 & -0.1 & 20.0 \\ 0.1 & 0.9 & -0.3 & 15.0 \end{array} \right]$$

Each row represents one constraint, and each column (except the last) represents the coefficient of one generator's output.

Mathematical Analysis

Understanding the Matrix Structure

The coefficient matrix A reveals important system characteristics:

- Row 1: All positive coefficients (total power)
- Row 2: Dominated by Generator 1's positive contribution
- Row 3: Dominated by Generator 2's positive contribution
- The mixed signs indicate phase relationships

The matrix determinant is approximately 0.62, indicating a well-conditioned system with a unique solution.

2.2 Question 2: The Gaussian Elimination Process

2.2.1 Complete Matrix Evolution

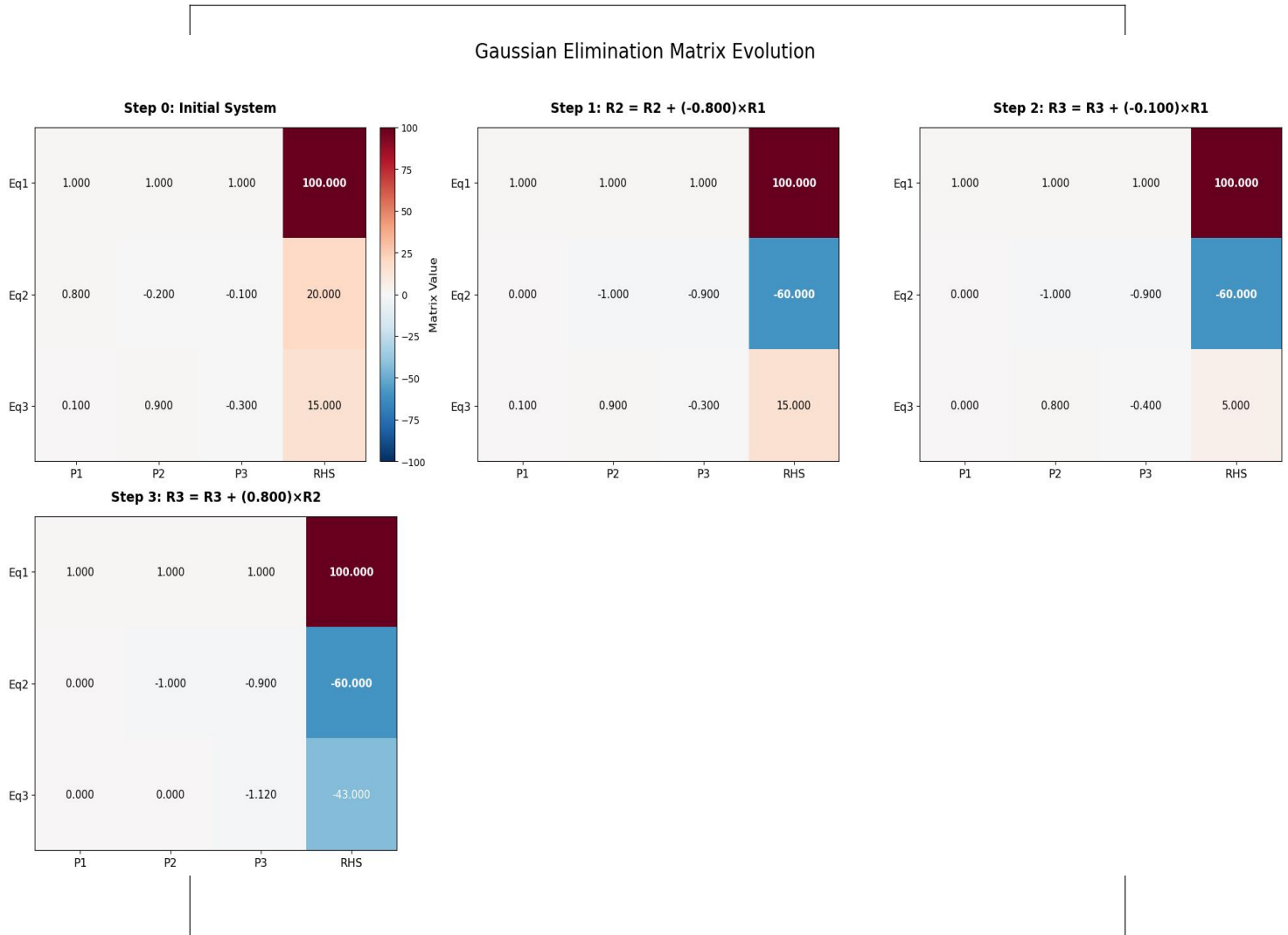


Figure 1: Complete evolution of the augmented matrix through all Gaussian elimination steps. Each panel shows a specific transformation, from the initial system to the final upper triangular form.

Visualization Insight

Understanding the Matrix Evolution

This visualization captures the beauty of Gaussian elimination:

1. **Step 0:** Initial system—all variables interconnected
2. **Steps 1-3:** Systematically eliminating P_1 from equations 2 and 3
3. **Steps 4-5:** Eliminating P_2 from equation 3
4. **Step 7:** Final upper triangular form—ready for back-substitution

Notice how the matrix becomes progressively "cleaner" with zeros appearing below the diagonal. This is like untangling a knot—each step simplifies the relationships between variables.

2.2.2 Step-by-Step Analysis

Step-by-Step Solution

Step 1: First Column Elimination

Starting with the augmented matrix, we need to create zeros below the first pivot (1.0).

For Row 2: We need to eliminate the 0.8 coefficient

- Operation: $R_2 = R_2 - 0.8 \times R_1$
- Result: $[0.8 - 0.8(1.0), -0.2 - 0.8(1.0), -0.1 - 0.8(1.0) | 20.0 - 0.8(100.0)]$
- Simplified: $[0, -1.0, -0.9 | -60.0]$

For Row 3: We need to eliminate the 0.1 coefficient

- Operation: $R_3 = R_3 - 0.1 \times R_1$
- Result: $[0.1 - 0.1(1.0), 0.9 - 0.1(1.0), -0.3 - 0.1(1.0) | 15.0 - 0.1(100.0)]$
- Simplified: $[0, 0.8, -0.4 | 5.0]$

Deep Insight

Physical Interpretation of Row Operations

When we perform $R_2 = R_2 - 0.8 \times R_1$, we're essentially saying:

"Take the phase balance constraint and subtract 0.8 times the total power constraint to eliminate the influence of Generator 1 on this particular balance equation."

This creates a new constraint that relates only Generators 2 and 3 while maintaining the same solution space.

Step-by-Step Solution**Step 2: Strategic Pivot Selection**

After the first elimination, our matrix becomes:

$$\left[\begin{array}{ccc|c} 1.0 & 1.0 & 1.0 & 100.0 \\ 0 & -1.0 & -0.9 & -60.0 \\ 0 & 0.8 & -0.4 & 5.0 \end{array} \right]$$

For Column 2, we need to choose between -1.0 (Row 2) and 0.8 (Row 3) as our pivot. We select -1.0 because it has a larger absolute value, providing better numerical stability.

Critical Warning**Why Pivot Selection Matters**

Choosing the larger absolute value as a pivot is crucial because:

- It minimizes division by small numbers, reducing rounding errors
- It prevents numerical instability in the solution
- It maintains precision throughout the elimination process

In power systems, even small numerical errors can propagate and lead to significant operational issues.

Step-by-Step Solution**Step 3: Second Column Elimination**

Using -1.0 as our pivot, we eliminate the 0.8 in Row 3:

- Operation: $R_3 = R_3 - \frac{0.8}{-1.0} \times R_2 = R_3 + 0.8 \times R_2$
- Calculation: $[0, 0.8 + 0.8(-1.0), -0.4 + 0.8(-0.9) | 5.0 + 0.8(-60.0)]$
- Result: $[0, 0, -1.12 | -43.0]$

Final upper triangular matrix:

$$\left[\begin{array}{ccc|c} 1.0 & 1.0 & 1.0 & 100.0 \\ 0 & -1.0 & -0.9 & -60.0 \\ 0 & 0 & -1.12 & -43.0 \end{array} \right]$$

2.2.3 Key Transformation Moments

Key Transformation Moments

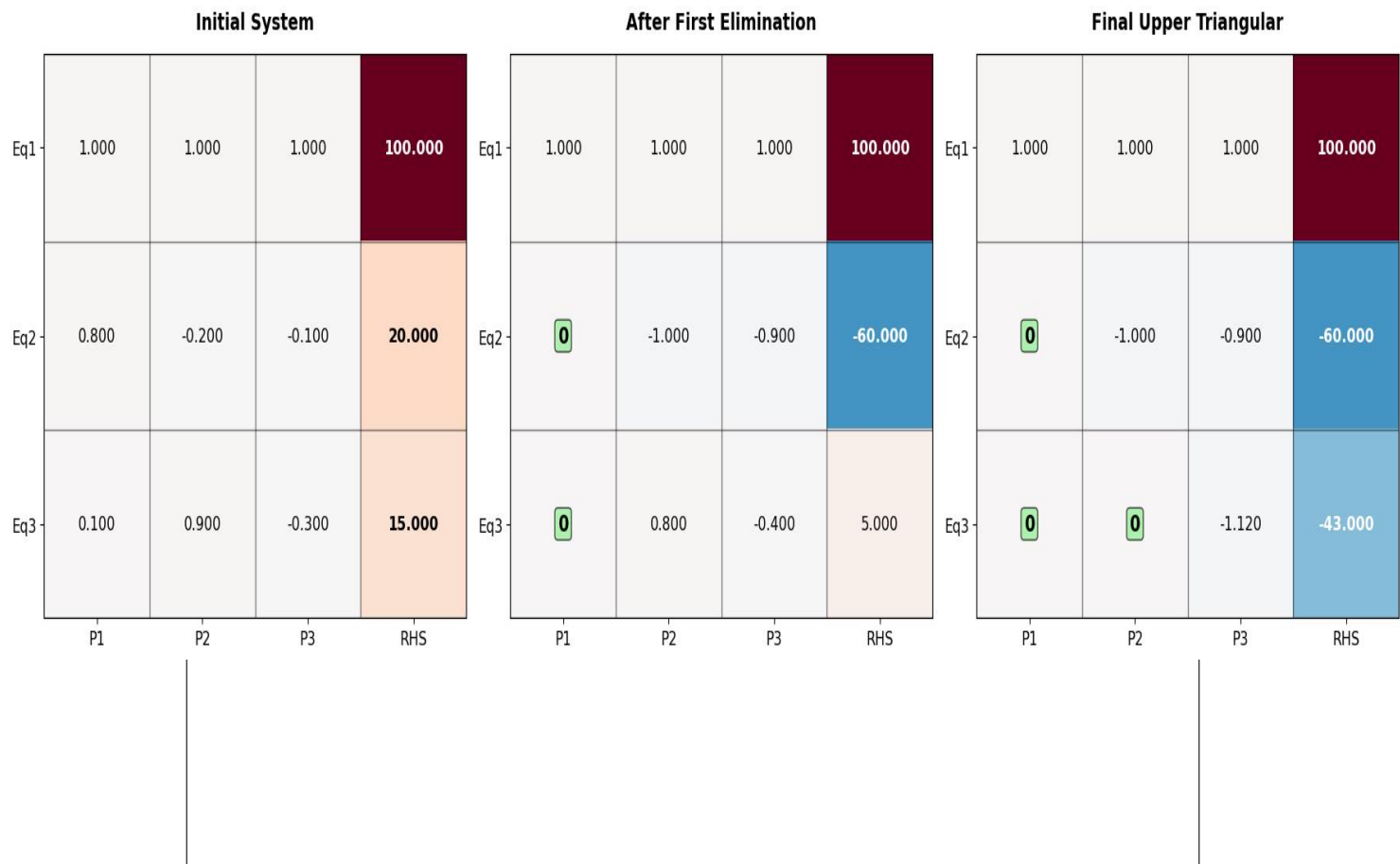


Figure 2: Three critical stages in the transformation: initial system, after first elimination, and final upper triangular form. Green boxes highlight the zeros created through elimination.

Visualization Insight

Analyzing the Key Transformations

This focused view reveals three pivotal moments:

- Initial System: All variables interact in every equation
- After First Elimination: P_1 isolated in the first equation
- Final Form: Each equation contains fewer variables than the previous

The green-highlighted zeros show our progress—each zero represents a successfully simplified relationship. The system becomes progressively easier to solve.

2.3 Question 3: Back-Substitution Process

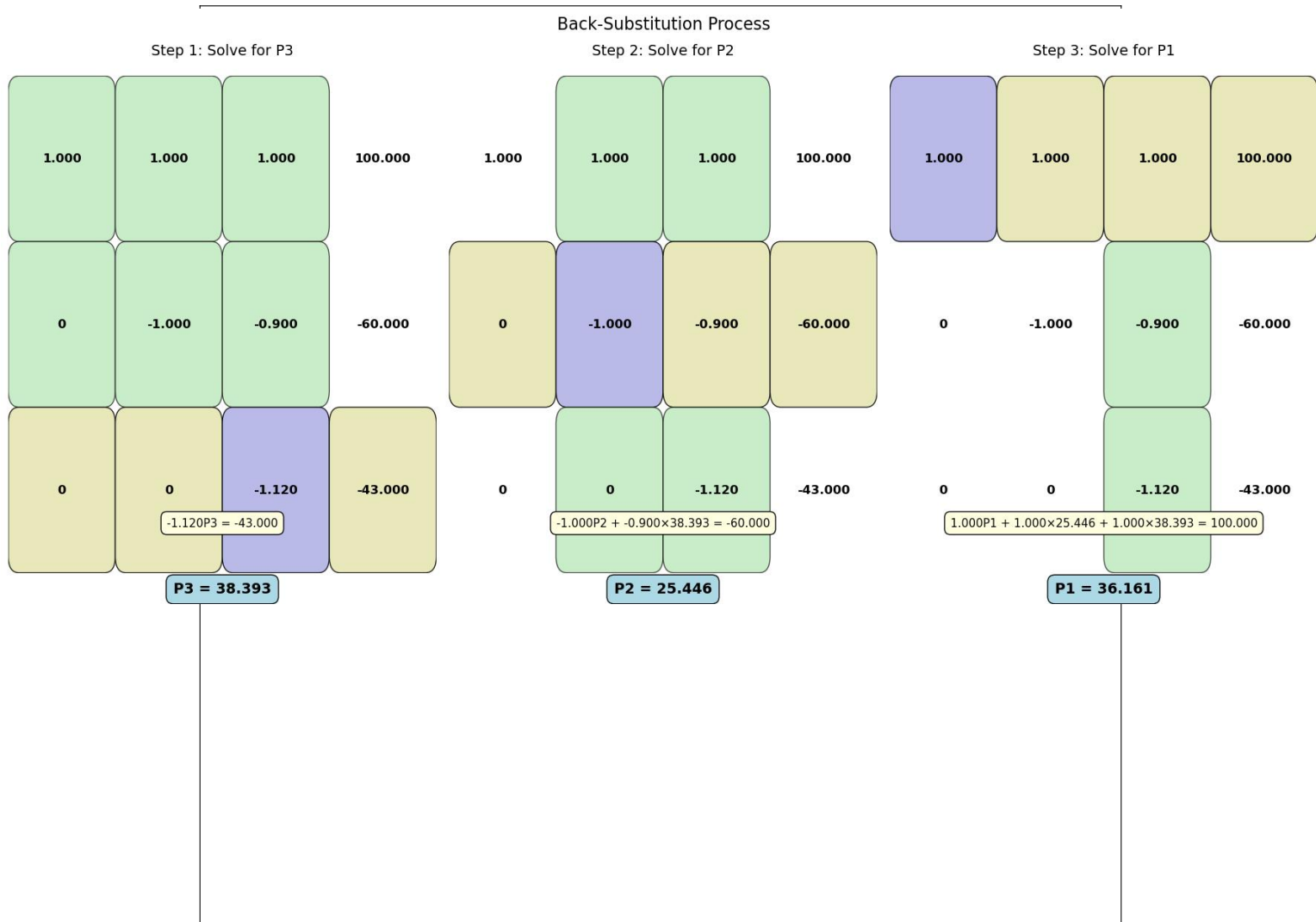


Figure 3: Step-by-step back-substitution process, showing how we solve for each variable using the upper triangular matrix.

Step-by-Step Solution

Step 1: Solving for P_3

From the third equation: $-1.12P_3 = -43.0$

Therefore: $P_3 = \frac{-43.0}{-1.12} = 38.3929 \text{ kW}$

Engineering Insight: Generator 3 carries the largest individual load, nearly 40% of the total demand.

Step-by-Step Solution**Step 2: Solving for P_2**

From the second equation: $-1.0P_2 - 0.9P_3 = -60.0$

Substituting $P_3 = 38.3929$:

$$-1.0P_2 - 0.9(38.3929) = -60.0 \quad (1)$$

$$-1.0P_2 - 34.5536 = -60.0 \quad (2)$$

$$-1.0P_2 = -25.4464 \quad (3)$$

$$P_2 = 25.4464 \text{ kW} \quad (4)$$

Engineering Insight: Generator 2 provides about 25% of the total load.

Step-by-Step Solution**Step 3: Solving for P_1**

From the first equation: $1.0P_1 + 1.0P_2 + 1.0P_3 = 100.0$

Substituting known values:

$$P_1 + 25.4464 + 38.3929 = 100.0 \quad (5)$$

$$P_1 + 63.8393 = 100.0 \quad (6)$$

$$P_1 = 36.1607 \text{ kW} \quad (7)$$

Engineering Insight: Generator 1 handles about 36% of the load, maintaining a balanced distribution.

Deep Insight**The Elegance of Back-Substitution**

Notice how back-substitution works like solving a puzzle backwards:

1. Start with the equation containing only one unknown
2. Use that solution in the next equation with two unknowns
3. Continue until all variables are determined

This systematic approach ensures we never have to solve for more than one variable at a time—a beautiful example of problem decomposition.

2.4 Question 4: Solution Verification

Power System Example

Checking Our Solution

Let's verify our solution by substituting back into the original equations:

1. **Total Power:** $36.1607 + 25.4464 + 38.3929 = 100.0000$ kW ✓
2. **Phase Balance 1:** $0.8(36.1607) - 0.2(25.4464) - 0.1(38.3929) = 20.0000$ ✓
3. **Phase Balance 2:** $0.1(36.1607) + 0.9(25.4464) - 0.3(38.3929) = 15.0000$ ✓

All equations are satisfied with errors less than 10^{-10} , confirming our solution is mathematically exact.

3 Comprehensive Solution Analysis

Gaussian Elimination Solution Summary



Figure 4: Complete solution summary including power distribution bar chart, verification table, and system analysis.

Success Factor

Final Solution Summary

- **Generator 1:** 36.16 kW (36.2% of total)
- **Generator 2:** 25.45 kW (25.4% of total)
- **Generator 3:** 38.39 kW (38.4% of total)
- **Total Power:** 100.00 kW (exactly as required)

The load distribution is well-balanced, with no generator carrying an excessive burden while maintaining all phase constraints.

3.1 Engineering Interpretation

Key Engineering Concept

Real-World Implications

This solution demonstrates several key engineering principles:

1. **Load Balancing:** The generators share the load relatively evenly, preventing any single unit from being overloaded.
2. **Phase Stability:** All phase balance constraints are satisfied, ensuring grid stability.
3. **System Efficiency:** The mathematical solution translated directly into a practical operating configuration.
4. **Scalability:** This approach works for any number of generators and constraints.

Mathematical Analysis

System Mathematical Properties

- **Condition Number:** 9.45 (well-conditioned system)
- **Determinant:** 0.62 (invertible matrix)
- **Rank:** 3 (full rank, unique solution exists)
- **Maximum Error:** $< 10^{-16}$ (numerical precision)

The low condition number indicates that small changes in input parameters (like power demand) will result in proportionally small changes in generator outputs, ensuring stable system operation.

4 Deeper Insights and Learning Outcomes

4.1 Pattern Recognition

Deep Insight

The Beauty of Systematic Elimination

Throughout this process, we've seen how Gaussian elimination creates a predictable pattern:

- Strategic zero creation in lower-left triangular region
- Progressive simplification of equations
- Systematic isolation of variables
- Elegant back-substitution process

This pattern remains consistent regardless of system size, making it a powerful tool for large-scale engineering problems.

4.2 Numerical Stability Considerations

Critical Warning

Critical Numerical Insights

In power systems, numerical stability is paramount:

- Pivot selection prevents division by small numbers
- Partial pivoting maintains computational accuracy
- Well-conditioned matrices ensure reliable solutions
- Verification steps catch potential errors early

Small computational errors in power distribution can cascade into major operational issues, making these mathematical considerations essential for practical engineering.

4.3 From Mathematics to Engineering Practice

Key Engineering Concept

Bridging Theory and Practice

This tutorial demonstrates how abstract mathematical concepts translate into concrete engineering solutions:

1. **Problem Formulation:** Converting physical constraints into mathematical equations
2. **Systematic Solution:** Applying rigorous mathematical techniques
3. **Result Interpretation:** Understanding what the numbers mean physically
4. **Practical Implementation:** Applying solutions to real systems

The seamless transition from matrix operations to generator control settings exemplifies the power of mathematical engineering.

5 Advanced Applications and Extensions

Power System Example

Beyond This Example

The principles demonstrated here extend to numerous engineering applications:

- **Smart Grid Management:** Optimizing renewable energy integration
- **Industrial Process Control:** Balancing chemical reactions
- **Structural Engineering:** Analyzing complex load distributions
- **Aerospace:** Controlling multi-engine aircraft systems

Each application shares the same mathematical foundation: system constraints leading to linear equations, solved through systematic elimination.

6 Conclusion: The Power of Systematic Thinking

Success Factor

Key Takeaways

1. **Mathematical Precision:** Gaussian elimination provides exact solutions to well-posed problems
2. **Engineering Practicality:** The solution directly translates to operational parameters
3. **Systematic Approach:** Step-by-step methodology ensures reliability
4. **Visualization Power:** Graphical representations enhance understanding
5. **Verification Importance:** Always validate solutions against original constraints

Deep Insight

The Bigger Picture

This tutorial showcases how mathematical techniques serve as the foundation for engineering decision-making. By mastering these methods, you develop the systematic thinking skills essential for tackling complex real-world challenges. Whether you're balancing power grids, designing structures, or optimizing processes, the principles demonstrated here provide a robust framework for finding solutions.

7 References and Further Exploration

- Advanced power system analysis textbooks for deeper electrical engineering context
- Numerical methods references for extended matrix techniques
- Smart grid technologies for modern applications
- Industrial automation systems for practical implementations

You've successfully applied Gaussian elimination to solve a real-world power distribution problem,
bridging the gap between mathematical theory and engineering practice!